Downloaded from gsabulletin.gsapubs.org on October 21, 2010

Geological Society of America Bulletin

Paleomagnetism and late diagenesis of Jurassic carbonates from the Jura Mountains, Switzerland and France

R.J.E. JOHNSON, R. VAN DER VOO and W. LOWRIE

Geological Society of America Bulletin 1984;95;478-488
doi: 10.1130/0016-7606(1984)95<478:PALDOJ>2.0.CO;2

Email alerting services click www.gsapubs.org/cgi/alerts to receive free e-mail alerts when new articles cite this article

Subscribe click www.gsapubs.org/subscriptions/ to subscribe to Geological Society of America Bulletin

Permission request click http://www.geosociety.org/pubs/copyrt.htm#gsa to contact GSA

Copyright not claimed on content prepared wholly by U.S. government employees within scope of their employment. Individual scientists are hereby granted permission, without fees or further requests to GSA, to use a single figure, a single table, and/or a brief paragraph of text in subsequent works and to make unlimited copies of items in GSA's journals for noncommercial use in classrooms to further education and science. This file may not be posted to any Web site, but authors may post the abstracts only of their articles on their own or their organization's Web site providing the posting includes a reference to the article's full citation. GSA provides this and other forums for the presentation of diverse opinions and positions by scientists worldwide, regardless of their race, citizenship, gender, religion, or political viewpoint. Opinions presented in this publication do not reflect official positions of the Society.

Notes

Geological Society of America

THE GEOLOGICAL SOCIETY OF AMERICA

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

Paleomagnetism and late diagenesis of Jurassic carbonates from the Jura Mountains, Switzerland and France

R.J.E. JOHNSON
R. VAN DER VOO
W. LOWRIE
Department of Geological Sciences, The University of Michigan, Ann Arbor, Michigan 48109
Institut für Geophysik, ETH-Hönggerberg, CH-8093 Zurich, Switzerland

ABSTRACT

A crucial fold test for magnetizations measured in Bajocian through Kimmeridgian limestones from the Jura Mountain region shows that rare blue-gray limestones found in this sequence preserve a prefolding magnetization. In contrast, tan limestones, which predominate in this sequence, contain secondary magnetizations acquired after folding in the late Miocene. The postfolding magnetizations in tan limestones appear to be carried in goethite and, because reversed as well as normal polarity directions are present, the magnetization in these limestones is probably a post-Miocene, pre-Brunhes (>700,000 yr) chemical remanent magnetization (CRM). Field relations between tan and blue-gray limestones, which are present in some cases in the same rock unit (for example, the Hauptrogenstein), suggest that the tan limestones may have formed from alteration of blue limestones. Whereas goethite pseudomorphs after pyrite are found in tan limestones, the occurrence of pyrite in the blue limestones supports this model, indicating that goethite could have formed from oxidation of pyrite. Abundant meteoric water probably was available for such an oxidation process during uplift of these strata subsequent to deformation.

Natural remanent magnetization in blue limestones appears to be carried in magnetite. The positive fold test for the characteristic magnetization in these rocks is significant at the 99% confidence level (k values for 24 site-mean directions before and after tilt correction are 9.3 and 43.2, respectively). The simplest interpretation for the origin of observed remanence in magnetite in the blue limestones is a primary magnetization of depositional or biogenic origin. Characteristic directions in these rocks yield a mean paleopole position of 77.7N, 148.4E (K = 26.0, A_{95} = 5.9). This pole is in reasonable agreement with one other Upper Jurassic pole from western Europe and with other Middle to Upper Jurassic poles from the Atlantic-bordering continents after rotation of these continents to a reconstructed configuration for Middle to Late Jurassic time.

INTRODUCTION

A paleomagnetic pole position for the Jurassic of western Europe is poorly known (Schott and others, 1981). A paucity of reliable paleomagnetic data for rocks of approximately the same age from North America also exists, as noted by McCabe and others (1982), and therefore there is uncertainty in the apparent polar wander history of the entire northern continent during the mid-Mesozoic. Limestones from the Jura Mountain region of eastern France and northwestern Switzerland were studied principally in an attempt to fill this notable gap in the Mesozoic paleomagnetic record for western Europe and to test the suitability of the rocks for magnetostratigraphy.

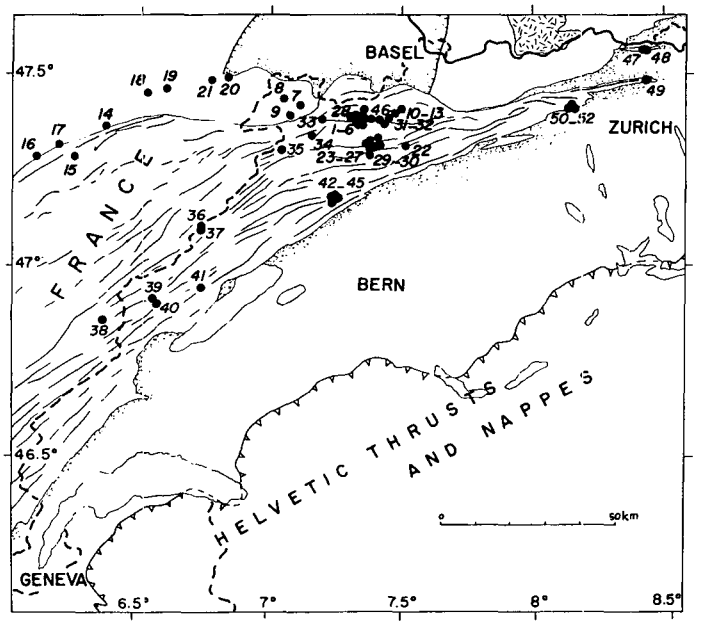


Figure 1. Jura Mountain region with numbered paleomagnetic sampling sites. Horizontal traces of fold axes in the Jura region are shown as discontinuous lines.

Limestones from the Jura region are representative of much of the Jurassic exposed in the stable part of western Europe. These rocks have natural remanent magnetizations (NRM) with very low intensities, making determination of reliable paleomagnetic directions from them difficult. This problem has largely prevented their use in paleomagnetic work in the past. However, cryogenic magnetometers that are now available allow the complete laboratory demagnetization of NRM in carbonate rocks necessary for directional analysis (Lowrie and others, 1980), and using this type of instrument it was possible to examine paleomagnetic directions in the Jura limestones. Identification of magnetic mineral components in carbonate rocks using conventional microscopic methods also can be a difficult problem, but with reliable rock magnetic tests it is possible to infer the magnetic mineralogy (Lowrie and Heller, 1982). The stabilities and probable modes of origin for observed magnetic remanence in carbonate rocks are inferred from an analysis of both NRM and rock-magnetic data.

Geological Society of America Bulletin, v. 95, p. 478–488, 12 figs., 5 tables, April 1984.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

In spite of their low-intensity magnetizations, limestones from the Jura region otherwise present good material for a paleomagnetic study, because (1) the Jurassic sequence here contains fairly uniform lithologies, well exposed over a broad area, (2) the Jurassic strata are well dated paleontologically, and (3) these strata are folded so that a field test for the age of magnetization is possible. The last consideration is particularly important because on the basis of previous studies it can be anticipated that Jurassic directions are fairly similar to present-day field directions. We wanted to avoid a possible misidentification of the age of magnetization by use of Graham's (1949) fold test.

An unexpected result of our study has been the evidence for a late diagenetic event in the Jura, as recognized in the often complex magnetizations. Although we can only speculate about this diagenesis insofar as the iron oxides are concerned, the effects appear to be quite pervasive. Our study simultaneously demonstrates the usefulness of paleomagnetism for dating such diagenetic events and emphasizes the dangers of misidentifying the age of magnetization in less favorable structural-sedimentological settings.

We have been fortunate in that a very small percentage of the Jura outcrops appear to have escaped diagenetic alteration, such that these rocks have yielded what we interpret to be a primary Jurassic paleopole. Given that marine magnetic anomalies for the Jurassic are not well defined, this paleopole will be useful for plate-tectonic reconstructions for the first opening phases of the Atlantic Ocean.

GEOLOGY

The Jura Mountains form a slightly arcuate belt of deformed sedimentary strata trending approximately northeast-southwest in Switzerland and eastern France (Fig. 1). Strata within the deformed belt are of Middle Triassic age and younger. The structural setting of the Jura has long been a classical example of a foreland fold-thrust belt overlying a décollement surface associated with Triassic evaporites (Rutten, 1969).

Middle and Upper Jurassic strata are widely exposed both in the folded Jura and in the Jura tableland bordering the Jura Mountains on the northwest. These strata were deposited as marine shallow-water sediments. Predominant marls in the Middle Jurassic reflect a supply of fine-grained terrigenous detritus during this time. In the Upper Jurassic, a variety of carbonate rock types records the development of a south-facing carbonate platform with organic reefs (Bollinger and Burri, 1970). Coral reefs had an important influence on sedimentation in the Jura until cessation of marine conditions in the latest Jurassic (Wilson, 1975). Associated ammonite faunas in these strata allow precise biostratigraphical correlation across facies boundaries (Gygi, 1969; Gignoux, 1955). Detailed biostratigraphical information in this region also demonstrates a progressive migration of lithofacies to the southeast in the Upper Jurassic (Gignoux, 1955), and lithologic units within the Oxfordian-Kimmeridgian are diachronous.

Predominant rock units in the Jura of interest to this study are bedded limestones, including Bajocian calcarenitic-oolitic lime sands (Hauptrogenstein), Callovian marls, and fine-grained limestones in the Oxfordian-Kimmeridgian that make up both fore-reef and back-reef facies. The coral-reef facies were not sampled because of their often coarse grain sizes. The fore-reef facies is represented by marls with discontinuous beds of microbioclastic lime silts (Terrain à Chailles) and micritic limestones with varying amounts of clay (for example, Effinger beds). The back-reef facies is characterized by chalky limestones and fairly pure micrites containing algal material (for example, Vorbourg beds). A few fine-grained crystalline limestone units within the uppermost Jurassic (Portlandian) were also studied, but this extremely variable section of abundantly gypsiferous

SERIES	STAGE	REMAGNETIZED SITES	"GOOD" SITES	FORMATION NAME
MALM	PORTLANDIAN		*7	"PORTLANDIAN"
		20		"J8a"
	KIMMERIDGIAN	13	9	"Ki"
		1,12,18,37	43	"HAUTES ROCHES"
	OXFORDIAN	10	42	"HAUPTMUMIENBANK"
		11,14,21,26	*8,19,36,49 39,40,41,44,45	"VORBOURGKALK"
		38,50,51	47,48,52	"VELLERAT", "EFFINGER AND GEISSBERG SCHICHTEN"
		22,23,24		"BIRMENSDORFER SCHICHTEN"
		27,28,34	6,17,30,31,32 33,46	"PICHOUX KALKE"
				"TERRAIN À CHAILLES"
DOGGER	CALLOVIAN			
	BATHONIAN	3,5,15,16	35	"PIERRE BLANCHE"
	BAJOCIAN	2,4,25,29		"HAUPTROGENSTEIN"
	AALENIAN			
LIAS	TOARCIAN			
	PLIENSBACHIAN			
	SINEMURIAN			
	HETTANGIAN			

Figure 2. Stratigraphic section sampled in the Jura region. Remagnetized sites are in tan limestones, and "good" sites are in blue-gray limestones; the asterisk denotes the occurrence of mixed-polarity magnetic directions.

shallow-water marine and brackish-water limestones was not extensively sampled. The total thickness of the Middle to Upper Jurassic section in the study area ranges from 250 to 300 m; absolute ages for these strata range from 170 Ma (Bajocian) to about 145 Ma (upper Kimmeridgian) (Van Eysinga, 1975).

The sedimentary rock units exposed within the Jura Mountains have been deformed into major fold structures. The trend of major folds follows the general arcuate shape of the Jura (Fig. 1), although individual folds are largely nonparallel and discontinuous (Laubscher, 1982). In the northeast Jura, folds trend nearly east-west; in the central Jura, they trend northeast-southwest; and in the south, they trend nearly north-south. Tightly compressed folds in the interior (southeast) part of the fold belt give way to more open folds with lower-limb dips in the northwest part, whereas Mesozoic strata exposed in the Jura tableland are nearly flat-lying. South-dipping thrust faults are numerous in the northeast Jura but are not developed elsewhere. Vertically dipping wrench faults with small displacements occur throughout the fold belt. Penetrative deformation of the rocks is in general confined to the rather sharply defined hinge regions of most folds. Deformation of the Jura region occurred in the latest Miocene, subsequent to the main phase of folding and nappe formation in the Alps (Laubscher, 1961). Although décollement during compression probably occurred in part of the Jura (Pierce, 1966), significant displacement (for example, rotation) relative to underlying basement of the northwest part of the fold belt and especially of the Jura tableland is highly unlikely.

SAMPLING

Paleomagnetic samples of Jurassic strata were obtained during two successive collecting seasons (1978 and 1979) by Van der Voo and Low-

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

rie. All sampling was done using a portable gas-powered core drill, and cores were oriented with magnetic compass and inclinometer. A few beds, usually representing a vertical thickness of 1 to 2 m, were drilled at each site.

A simplified stratigraphic column for the Jura section (Fig. 2) shows the formations that were sampled and their approximate ages. Almost all of the Jurassic limestones occurring in the Jura region are tan colored, but in a few exposures of the bedded units blue-gray colored limestones were observed. For example, some beds of the oolitic Hauptrogenstein show a bluish interior rimmed by tan limestone. This same relationship is also seen in some of the "Chailles" and in the Oxfordian-Kimmeridgian micrites, although it is very infrequent (<1%). Blue-gray and tan limestones have similar predominantly massive textures; however, relatively unaltered sulphides are present only in the blue limestones as cubic and framboidal pyrite. Pyrite crystals are occasionally altered in these blue limestones and are indicated by a zone of light gray to yellowish discoloration a few times the diameter of the grain. This color variation in the Jura limestones is a feature that has rarely been noted previously, and the tan color was interpreted as the original color of these rocks (A. Bürgin, 1978, personal commun.). Most sites of a first collection of samples were in tan limestones, but a few sites in blue limestones from the Hauptrogenstein and from gently dipping Terrain à Chailles were also included. The preliminary results from laboratory measurements done on this collection in Zurich showed significant differences between tan and blue limestones in magnetic properties and directions. Additional sites in blue limestones, especially from folded beds, were required to confirm the initial results and to complete the fold tests. A number of sites eventually found in blue limestones in the folded "Chailles" and Oxfordian-Kimmeridgian micrites constitute the second collection (sites 33–52 in Figs. 1 and 2).

Cognizant of the possibilities of *in situ* rotations of some parts of the fold belt, we have extended our sampling along strike in the folded Jura as well as into the more flat-lying part of the foreland (Jura tableland) in order to test for declination variations indicative of rotations (for example, Fig. 1). However, sampling was confined to parts of the Jura that have not been extensively faulted.

LABORATORY METHODS AND DATA ANALYSIS

Paleomagnetic Methods

Standard 2.54-cm-diam paleomagnetic core samples were cut into specimens with length 2.2 cm for paleomagnetic measurements (specimen vol = 10 cm^3). The NRM of specimens was measured, and one specimen from each core was demagnetized stepwise to examine the vectoral components and the stability of the NRM. Demagnetization was performed either by applying alternating magnetic fields of progressively higher intensity to individual specimens (AF demagnetization), or by heating approximately ten specimens at a time to progressively higher temperatures in a zero-magnetic field. Magnetization of specimens was measured following each step of treatment, and the remanent magnetization removed by any treatment was resolved by simple vector subtraction. Demagnetization treatment sought to selectively remove individual components of magnetization (each acquired in an instant of geologic time) over different ranges of applied treatment. Demagnetization was continued in most cases until the remanence intensities reached 3×10^{-6} A/m, which corresponds to the noise level of the magnetometer, following a stepwise procedure that ensured that no component magnetizations remained undetected.

Preliminary demagnetization experiments done in Zurich and later work completed at the University of Michigan used similar instruments. Cyrogenic magnetometers used for magnetic measurements were three-axis (Zurich) and two-axis (University of Michigan) instruments made by Superconducting Technology (Goree and Fuller, 1975). The apparatus used for AF demagnetization at the University of Michigan was a commercial (Schonstedt GSD-1) instrument, capable of reaching 100 mT peak field. The sample was demagnetized in three orthogonal directions in the same peak field before being measured. At the ETH in Zurich, the sample was tumbled about two orthogonal axes while being demagnetized in alternating fields; a theoretical calculation shows that a given peak field applied in the Zurich apparatus is about 1.5 times more effective than the same peak field applied in the static method of the Schonstedt apparatus. In both laboratories, thermal demagnetizations were carried out in Schonstedt (TSD-1) furnace units in field-free space of less than 10 nT.

Analysis of NRM measurement data for each specimen was performed using orthogonal projections of the vector end points of magnetization remaining after each step of treatment (two projections representing projection onto the horizontal plane as well as onto a vertical plane completely describe the vector magnetization; "demagnetization diagrams" after Zijderveld, 1967). Progressive demagnetization produces a demagnetization trajectory defined by successive end points of the magnetization, and in this graphic representation, both the horizontal and the vertical projections of this trajectory are shown. Selective removal of an individual component of magnetization was identified from visual inspection of a demagnetization diagram, as a linear segment of a trajectory determined by at least four data points. By these criteria, AF demagnetization consisting of approximately ten treatment steps with a progressive increase in applied AF intensity in 5 mT increments was effective in isolating multiple component-magnetizations in most specimens. An anhysteretic component of magnetization (ARM) was occasionally acquired by specimens during AF treatment using the static method. However, the contribution of this laboratory-acquired magnetization was minimized by reversing the orientation of the sample in the demagnetizer, repeating the treatment, and averaging the results (French, 1976). Thermal treatment was discontinued after initial experiments produced large angular changes in magnetization direction between successive steps above 300 °C in all but a few of the specimens treated. This was attributed to spurious magnetizations acquired by newly formed magnetic minerals in a small background field during heating, which are ordinarily difficult to completely prevent but which profoundly reduce the signal-to-noise ratio in such weakly magnetized samples. Data for samples that acquired spurious magnetizations in the laboratory were not considered in any subsequent analysis. Directions (declination and inclination) were calculated for all component magnetizations that were reliably identified from demagnetization diagrams for specimens, and these were used to compute mean directions (site and formation means) and the variance in the distribution of individual directions about a mean (Fisher, 1953).

Rock-Magnetic Methods

After core specimens were demagnetized, a number of them were given an isothermal remanence magnetization (IRM) in steady magnetizing fields at laboratory temperatures of 25 °C; IRM was measured, and specimens were treated again at a higher field until maximum remanence (saturation IRM) was achieved or 5 T fields were reached. The magnetic field required to saturate specimens depends on the maximum coercivity grains present in the specimen. In that different ferromagnetic minerals have quite different coercivities, the saturating field measured in this way is useful for identifying the contained magnetic minerals (Lowrie and Heller, 1982; Dunlop, 1972). Specimens used for these experiments were unheated prior to magnetization. IRM in multiple specimens from the same

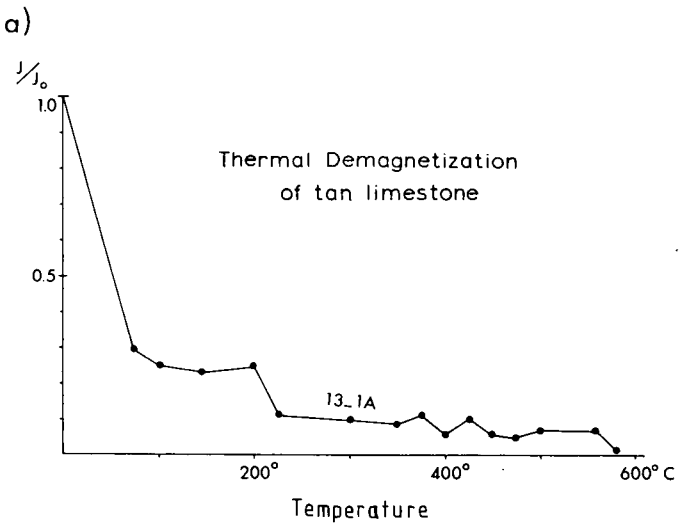
Downloaded from gsabulletin.gsapubs.org on October 21, 2010

TABLE 1. SITE-MEAN PALEOMAGNETIC DATA FOR THE TAN-COLORED MICRITES

Site number	$J_{av.} \pm$ s.d. (10^{-5} A m^{-1})	N/N_0	R/N	Before tectonic correction			After tectonic correction
				Decl/Incl deg	k	α_{95}	Decl/Incl deg
1	35.6 ± 75.2	14/14	6/8	184/ 61	26.2	7.9	188/ 68
2	1.3 ± 1.0	14/14	7/7	1/+64	3.4	25.4	184/+20
3	21.8 ± 59.6	11/11	1/10	353/+55	40.8	7.2	190/+64
4	2.7 ± 1.6	12/12	0/12	342/+67	6.1	19.1	211/+52
5	3.5 ± 0.5	13/13	0/13	352/+62	32.3	7.4	205/+43
10	1.6 ± 1.1	14/14	3/11	66/+65	4.5	21.2	167/+19
11	3.4 ± 2.9	16/16	0/16	351/+52	5.9	16.7	212/+49
12	3.2 ± 3.0	9/9	3/6	80/+80	3.1	34.8	174/+26
13	7.8 ± 7.1	14/14	3/11	12/+54	20.8	8.9	29/+70
14	1.7 ± 0.4	9/9	0/9	341/+60	30.6	9.5	338/+50
15	2.6 ± 1.2	11/11	0/11	354/+65	10.0	15.2	357/+32
16	3.4 ± 1.4	11/11	0/11	5/+68	34.2	7.9	23/+77
18	5.6 ± 1.5	11/11	0/11	359/+62	84.2	5.0	359/+62
20	5.2 ± 1.6	12/12	0/12	360/+65	29.1	8.2	356/+58
21	8.9 ± 5.1	9/9	0/9	352/+62	75.1	6.0	352/+62
24	8.1 ± 5.1	12/12	0/12	338/+52	18.0	10.5	132/+86

Note: Numbers correspond to those in Figure 1; $J_{av.} \pm$ s.d. represents the average magnetic intensity per site with its standard deviation; N/N_0 is the ratio of samples used in the statistical analysis to those demagnetized with alternating fields. R/N is the ratio of samples with reversed polarity to those with normal polarity; k and α_{95} are the statistical parameters associated with the site means.

site and often from the same bed was examined; each specimen was magnetized parallel to its core axis in a uniform direction using either a water-cooled electromagnet with a wide pole gap (to 1.3 T), or a cryogenic solenoid magnet available at the University of Michigan (to 5 T). Blocking temperatures of saturation IRM in specimens were later examined by stepwise thermal demagnetization. Additional experimental tests were performed on a portion of the limestones that were suspected of containing magnetite, in an attempt to investigate the magnetic-domain state of the magnetite. These tests involved (1) examining the effect of cooling samples in a zero-magnetic field below 130 °K, the magnetocrystalline anisotropy transition temperature for magnetite (Ozima and others, 1964); and (2) examining the stability with respect to AF demagnetizing fields of an induced saturation ARM (ARMs) relative to the stability of an IRMs in the same samples (Lowrie-Fuller test). Above 130 °K, the magnetocrystalline preferred axes of magnetization are the [111] directions, whereas below this temperature the "easy" axes of magnetization are the [100] directions. Previous work using both synthetic magnetite (Ozima and others, 1964) and natural magnetite-bearing samples (Merrill, 1970) showed that upon cooling or warming through 130 °K, multidomain samples will experience a loss of remanence, whereas little loss of remanence is expected for single-domain magnetite grains (Kobayashi and Fuller, 1968). This is true because for single-domain grains, small depar-



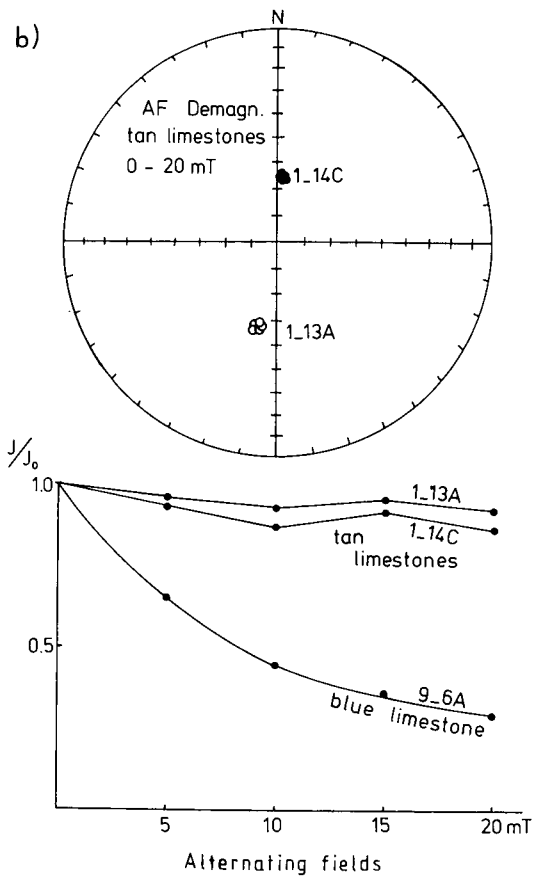


Figure 3. Decay of NRM in tan-limestone samples during stepwise thermal (a) and AF (b) demagnetization (J/J_0 is intensity normalized to the initial intensity). An equal-area plot in diagram b shows the directions of magnetization remaining in two tan-limestone samples after each successive step of AF treatment (filled circles are in lower hemisphere, and open circles are in upper hemisphere). AF demagnetization of NRM in a blue-limestone sample is also shown in diagram b for comparison.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

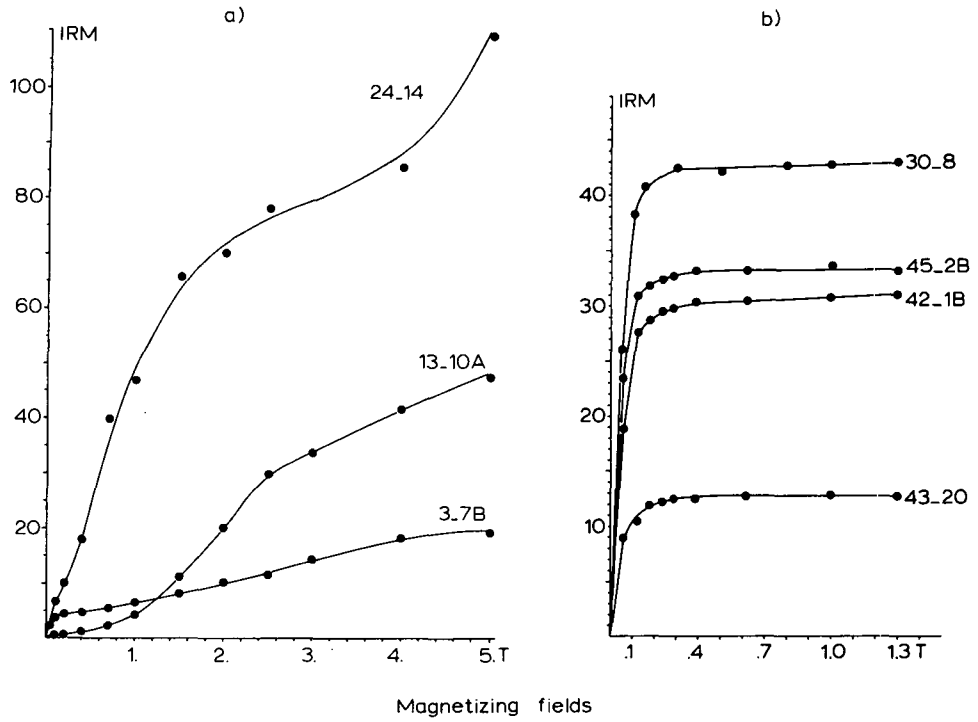


Figure 4. Acquisition of IRM in tan (a) and blue (b) limestones (units along vertical axes are in 10^{-3} A/m).

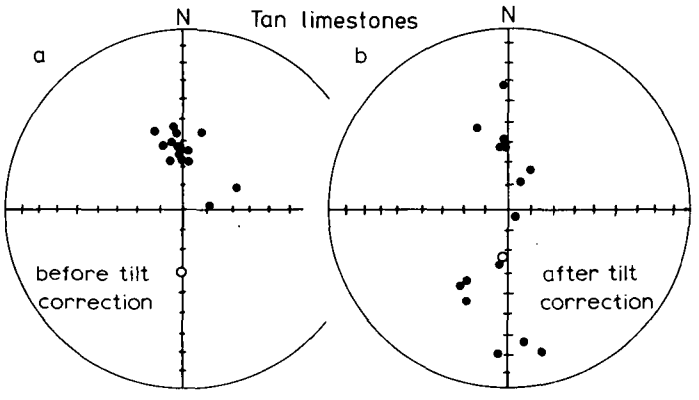


Figure 5. Equal-area plots of site-mean directions remaining in tan limestones after treatment to 20 mT, before tilt correction (a) and after tilt correction (b).

tures from a spherical shape will cause shape anisotropy to completely dominate all other anisotropies. Grains with a small number of magnetic domains (0–2) and intermediate in size (diameter $\approx 0.2 - 2\ \mu m$) between single-domain and true multidomain grains (called pseudo–single-domain) also exhibit only a very small loss of remanence when cycled through a low-temperature transition (Levi and Merrill, 1976). A test based on this observed behavior may be used to estimate the amount of remanence controlled by magnetocrystalline anisotropy energy. The behavior (relative stability) during AF demagnetization of ARM and IRM in samples can yield similar information. ARM in synthetic single-domain size samples consistently shows a greater stability than IRM to AF treatment, whereas the reverse relationship holds in synthetic multidomain samples. However, application of this test to samples containing a range of grain sizes or to natural-magnetite–bearing samples may produce less easily interpreted results (Johnson and others, 1975). In the experiments conducted here, a saturation ARM was given to core samples along their axis using the Schonstedt AF apparatus in a steady biasing field of 0.18 mT. A steady field parallel to the applied AF field was produced by large Helmholtz coils surrounding the Schonstedt solenoid. Cores were demagnetized in nine or ten steps using progressively higher alternating fields (with sample in zero-magnetic field) to 100 mT. An ARMs was given to each sample a second time and its magnetization was measured at 25 °C before and after the sample was twice cycled between room temperature and -196 °C (while in zero-magnetic field). The ARM remaining in samples after low-temperature treatment was subsequently removed by progressive AF treatment. IRMs (at 25 °C) was given to these samples after the low-temperature experiments had been completed.

PALEOMAGNETIC AND ROCK-MAGNETIC RESULTS

Tan Limestones

NRM intensities measured in tan limestone samples were found to be variable (Table 1), but the response of most of these samples to laboratory demagnetization was quite uniform. Thermal demagnetization is characterized by rapid decay of the NRM toward the origin with most of the intensity removed by 150 °C (Fig. 3a). Alternating field treatment, on the other hand, did not significantly affect the NRM in these rocks; the magnetization direction and intensity after treatment to 20 mT (using AF demagnetizer with specimen tumbler) are approximately the same as those of the NRM (Fig. 3b).

The high coercivity of the magnetic carrier in the tan limestones is also demonstrated by the acquisition of IRM in samples of that lithology. Figure 4a shows that tan limestone samples do not reach saturation in 5 T. After an initially steep rise, which indicates the presence of a small magnetite component, the acquisition curves are often concave upward. The combination of very high coercivity (Fig. 4a) and low blocking temperature (Fig. 3a) is typical for goethite, which we interpret to be the predominant carrier of remanent magnetization in the tan limestones. Comparable

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

TABLE 2. SITE MEAN PALEOMAGNETIC DATA FOR THE BLUE-GRAY MICRITES

Site number	$J_{av.} \pm$ s.d. (10^{-5} A m^{-1})	N/N_o	R/N	Before tectonic correction					After Tectonic correction		
				Decl/Incl deg	k	α_{95}	Pole position		Decl/Incl deg	Pole position	
6	21.6 ± 12.4	19/19	0/19	20/+55	105.1	3.3	71.0°N,	128.9°E	35/+64	66.1°N,	88.5°E
7	1.4 ± 1.0	15/15	8/7	216/−60	5.8	17.4	63.6°N,	98.1°E	216/−59	63.1°N,	100.4°E
8	11.4 ± 9.3	12/12	4/8	15/+59	51.1	6.1	76.8°N,	126.8°E	15/+59	76.8°N,	126.8°E
9	3.1 ± 0.2	12/12	0/12	36/+63	28.8	8.2	65.0°N,	90.5°E	15/+60	77.5°N,	123.0°E
17	2.7 ± 0.9	14/14	0/14	9/+65	102.8	3.9	83.9°N,	94.4°E	19/+65	77.1°N,	89.7°E
19	3.9 ± 1.1	8/8	0/8	4/+52	19.3	13.0	75.0°N,	173.2°E	3/+47	70.8°N,	178.3°E
22	20.8 ± 8.1	0/14	magnetic components not separated								
23	4.2	0/2	only two samples								
26	18.5 ± 7.2	0/12	magnetic components not separated								
27	11.8 ± 3.2	0/13	magnetic components not separated								
28	22.2 ± 10.7	0/12	magnetic components not separated								
30	23.2 ± 27.9	14/15	0/14	10/+73	59.3	5.2	77.3°N,	31.6°E	358/+47	70.8°N,	192.8°E
31	10.1 ± 2.5	12/12	0/12	249/+63	34.6	7.5	20.0°N,	322.3°E	18/+46	65.6°N,	146.0°E
32	21.3 ± 8.7	12/12	0/12	355/+65	103.5	4.3	86.6°N,	272.6°E	9/+54	75.5°N,	156.6°E
33	22.3 ± 13.6	8/11	0/8	341/+26	114.9	5.2	52.7°N,	219.5°E	312/+65	57.9°N,	295.3°E
34	7.1	0/2	only two samples								
35	2.3 ± 1.3	3/6	0/3	11/ 0	31.8	22.2	41.7°N,	172.3°E	30/+49	61.1°N,	123.3°E
36	19.6 ± 3.2	5/5	0/5	18/+68	92.3	8.0	77.5°N,	74.8°E	4/+61	83.3°N,	158.3°E
37	4.5 ± 1.1	0/6	unstable magnetizations								
38	42.8 ± 9.3	0/9	magnetic components not separated								
39	13.9 ± 3.3	4/4	0/4	19/+63	91.2	9.7	76.5°N,	100.1°E	0/+57	80.7°N,	186.6°E
40	34.4 ± 4.8	6/7	0/6	345/+29	58.6	8.8	56.0°N,	213.3°E	11/+52	72.9°N,	152.1°E
41	62.8 ± 25.2	6/8	0/6	336/− 4	32.9	11.9	36.9°N,	217.7°E	336/+61	72.1°N,	271.7°E
42	14.6 ± 6.7	8/11	0/8	54/+78	20.5	12.5	61.5°N,	78.9°E	7/+53	75.4°N,	163.6°E
43	5.7 ± 2.5	4/6	0/4	43/+66	29.6	17.2	56.3°N,	42.0°E	16/+44	65.1°N,	151.2°E
44	3.5 ± 2.0	5/6	0/5	74/+71	29.3	14.4	45.3°N,	58.1°E	15/57	75.4°N,	133.0°E
45	27.0 ± 14.4	6/6	0/6	352/+68	141.4	5.7	83.5°N,	116.3°E	1/+54	77.3°N,	183.6°E
46	12.6 ± 3.8	4/4	0/4	333/+79	256.8	5.8	64.8°N,	344.7°E	22/+35	56.7°N,	147.3°E
47	4.1 ± 2.5	6/6	0/6	7/+57	44.0	10.2	78.8°N,	158.7°E	356/+64	86.7°N,	245.4°E
48	4.0 ± 1.0	5/5	0/5	355/+52	72.9	9.0	74.6°N,	204.5°E	350/+55	75.9°N,	223.9°E
49	5.2 ± 0.7	5/5	0/5	352/+59	26.4	15.1	80.4°N,	228.1°E	22/+61	73.5°N,	109.6°E
50	5.3 ± 2.6	0/6	unstable magnetizations								
51	7.0 ± 3.3	0/7	unstable magnetizations								
52	5.9 ± 3.4	6/12	0/6	18/+31	30.5	12.3	55.8°N,	156.3°E	23/+47	63.7°N,	137.1°E

Note: explanation as for Table 1. N/N_o is the ratio of samples used in the statistical analysis to those demagnetized with alternating fields, except for sites 45 and 46, where three and two thermal demagnetizations, respectively, have been included.

conclusions were reached by Heller (1977, 1978) in extensive rock-magnetic and paleomagnetic studies of similar limestones from southern Germany.

Demagnetization treatment indicates a univectorial magnetization in tan limestones. Directions for magnetization remaining in tan samples after 20 mT are summarized by site in Table 1, and site-mean directions for this magnetization before and after applying a simple tilt correction are shown in Figure 5. The fold test is clearly negative, indicating that magnetizations in the tan limestones are postfolding in age. In that reversals are present among these postfolding directions (see Table 1), we argue that the goethite carries a post-Miocene, pre-Brunhes (>700,000 yr) chemical remanant magnetization rather than a viscous remanence (VRM).

Blue Limestones

NRM intensities measured in samples of blue limestone were generally similar to those measured in the tan limestones (Table 2). A large number of blue limestone samples, however, showed that, in contrast to magnetization in the tan limestones, both the direction and intensity of magnetization changed with application of AF treatment (for example, Fig. 3). Orthogonal demagnetization diagrams (Zijderveld, 1967) illustrate the main features of the NRM observed in blue limestones (Fig. 6). A "soft" magnetization constituting a large proportion of the NRM in virtually all blue limestone samples was removed in treatment to 25 mT or 250 °C. Despite occasionally noisy demagnetization trajectories, in many samples a soft component was determined that was approximately parallel to the present-day field direction (Fig. 6). A comparison of sample directions for soft components isolated in a number of samples from folded strata, before and after tilt correction, clearly shows the postfolding age for this magnetization (Fig. 7). In view of the exclusively normal polarity for this component, we suggest that it is a viscous remanent magnetization (VRM). Continued treatment of blue limestone samples above 25 mT or 250 °C resulted in progressive decrease of magnetization intensity but the change in direction of magnetization in most samples was small. Measurable magnetization in almost all samples was completely removed by 100 mT. Small magnetizations that remained after treatment to high alternating fields were observed in a few samples, but the occurrence of these was not systematic (that is, not found in all samples from a single site), and their directions were random. Given that magnetizations removed in treatment above ~25 mT or 250 °C, calculated by vector subtraction, yielded the most consistent directions at the site level (highest k), these are the characteristic directions reported for the blue limestones.

Table 2 presents a summary of characteristic directions for sites in the Oxfordian-Kimmeridgian blue limestones. Not listed are the data for those sites from which a characteristic direction could not be determined (sites 22, 26–28, 37, 38, 50, and 51). Although treatment of samples from sites 22 and 26–28 was continued to alternating fields as high as 40 mT, and magnetization in some samples appeared to approach a characteristic direction, stable end points were not achieved. In a similar fashion, detailed thermal and alternating field treatments of samples from site 38 yielded only curved demagnetization trajectories, indicating simultaneous removal of two or more components (Fig. 8a). Treatment of samples from sites 37, 50, and 51 in alternating fields to 25 mT produced no stable end points and this treatment removed all measurable magnetization in these samples (for example, Fig. 8b). On the other hand, there remain a large number of sites for which samples did yield characteristic directions (Fig. 9). We note the normal polarity of all sites except sites 7 and 8, which showed mixed polarities. From the data of Table 2, mean directions, mean poles, and their respective statistical parameters (Fisher, 1953) were calculated both before and after correction for structural tilt. A comparison of precision parameters for means (directions or poles) before tilt correction (k_1) and after tilt correction (k_2) shows considerable improvement in precision after

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

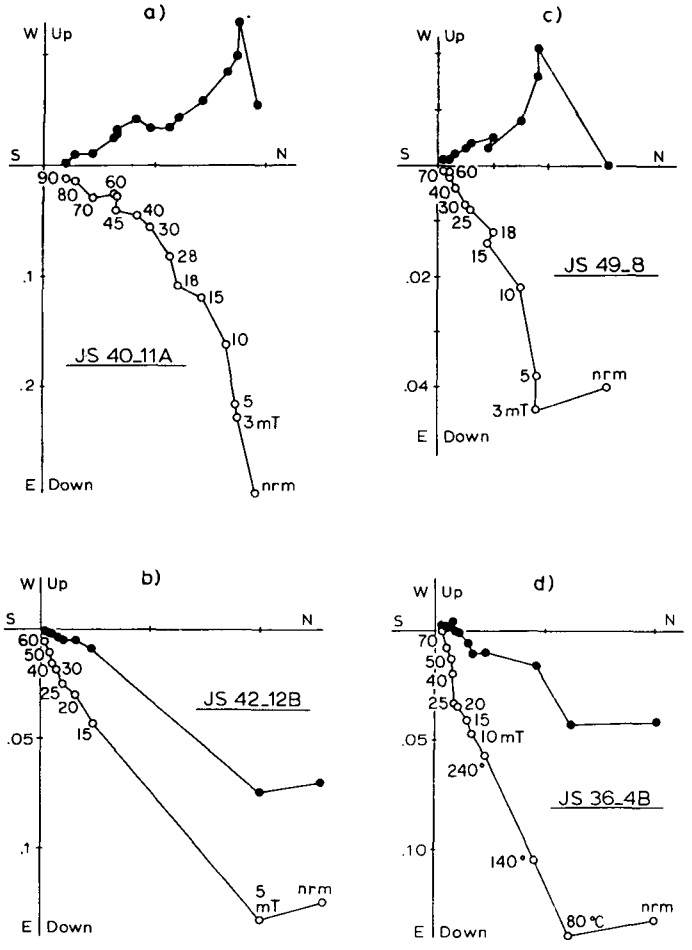


Figure 6. Orthogonal demagnetization diagrams (Zijderveld, 1967) for AF treatment (a–c) and thermal- AF treatment (d) of blue limestone samples. Co-ordinates are uncorrected for structural tilt. Solid circles are projections onto the horizontal plane, and open circles are projections onto the north-south vertical plane. Units along the axes are in 10^{-6} A/m.

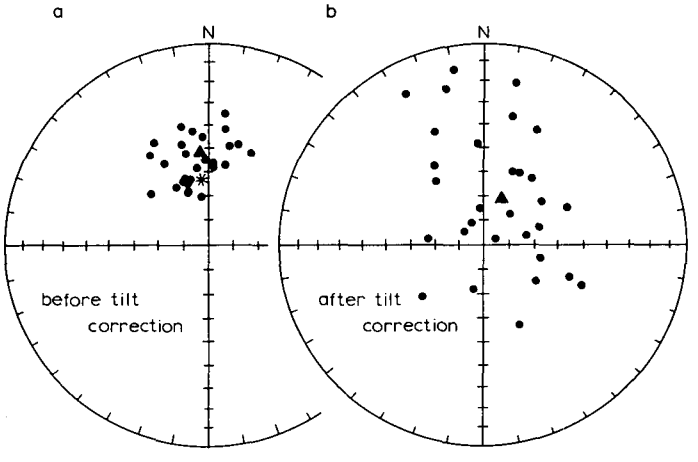


Figure 7. Equal-area plot of sample directions for soft components found in blue limestones before tilt correction (a) and after tilt correction (b).

tilt correction (Table 3), and the ratio k_2/k_1 for either mean directions or mean poles indicates a positive fold test significant at the 99% confidence level (McElhinny, 1964) as also shown in Figure 9. The characteristic magnetization in blue limestones thus is a prefolding magnetization.

Bluish samples from folded beds of the Hauptrogenstein (Bajocian age) (sites 25 and 29) that were examined with the first collection of limestones also yielded characteristic directions after 20 mT that tended to pass the fold test (Table 4). However, a fairly large scatter in directions corrected for structural tilt remained, and the positive fold test for these 21 samples from 2 sites is not significant. The imprecise grouping of directions for the Hauptrogenstein may be attributed to incomplete removal of secondary directions, or to a randomizing effect for magnetic directions of oolites. We note that, in full agreement with the results from the micrites, the tan alteration rims of the Hauptrogenstein tend to fail the fold test (Table 4).

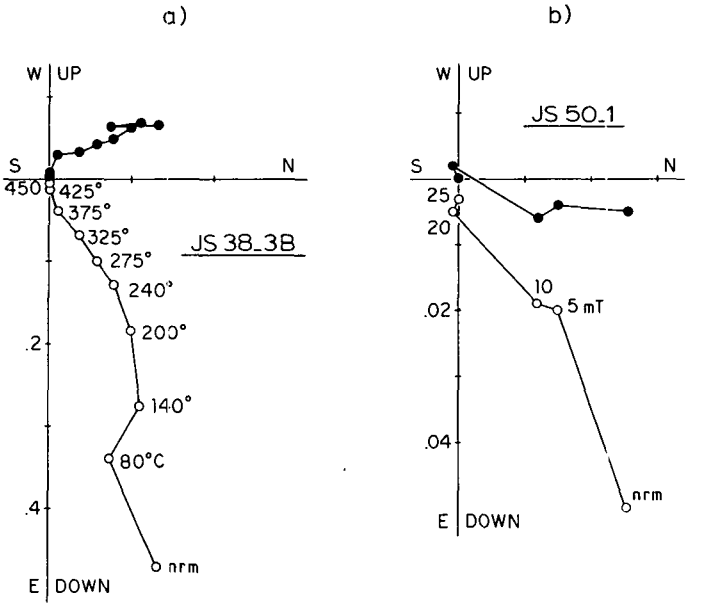


Figure 8. Orthogonal demagnetization diagrams (as in Fig. 6) for blue limestone samples that did not yield characteristic directions. Units along the axes are in 10^{-6} A/m.

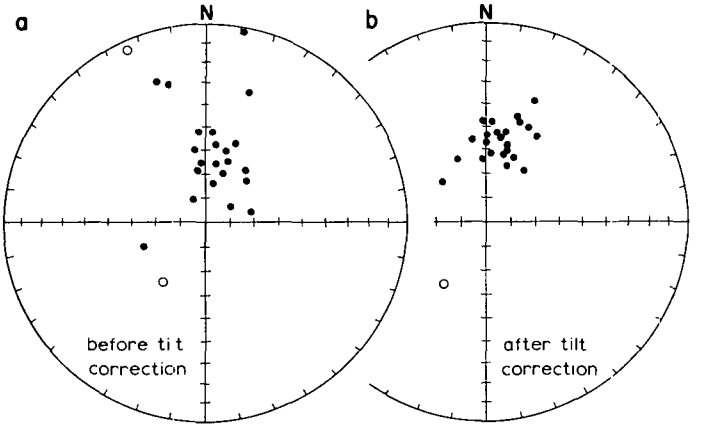


Figure 9. Equal-area plot of site means for characteristic directions in blue limestones before tilt correction (a) and after tilt correction (b).

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

TABLE 3. SUMMARY OF MEAN DIRECTIONS AND POLES

Lithology	N	Before tectonic correction					After tectonic correction					k_2/k_1
		Mean decl/Incl	k_1	α_{95}	Pole position		Mean decl/Incl	k_2	α_{95}	Pole position		
Tan micrites	16	359.1/+63.8	43.0	5.7	88.3°N,	210.8°E	215.3/+83.0	3.8	21.9			0.088
Blue micrites, site means	24	4.8/+58.2	9.3	10.3			10.0/+56.0	43.2	4.6	77.2°N,	149.0°E	4.650
Blue micrites, poles	24		7.8	11.3	83.1°N,	136.3°E		26.0	5.9	77.7°N,	148.4°E	3.330

Note: N is the number of entries (site means or poles) in the statistical analysis; Decl/Incl are declination and inclination, respectively, in degrees; k and α_{95} are the statistical parameters associated with the means; k_2/k_1 is the ratio of the precision parameters, after and before the tectonic correction, respectively.

TABLE 4. DIRECTIONS FROM THE HAUPTROGENSTEIN OOLITES

Lithology	N	Before tectonic correction			After tectonic correction			
		Decl/Incl	k_1	α_{95}	Decl/Incl	k_2	α_{95}	k_2/k_1
Blue layers	21	9/+69	3.2	21.0	13/+60	4.7	16.7	1.47
Tan rims	11	317/+57	4.8	23.3	333/+41	2.8	34.0	0.58

Note: N is the number of entries (samples) in the statistical analysis; Decl/Incl are declination and inclination, respectively, in degrees; k and α_{95} are the statistical parameters associated with the mean directions; k_2/k_1 is the ratio of the precision parameters, before and after tectonic correction, respectively.

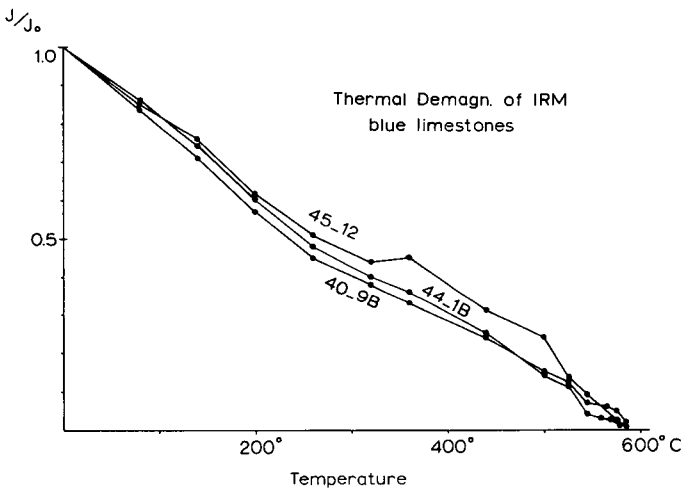


Figure 10. Thermal demagnetization of saturation IRM in blue limestone samples.

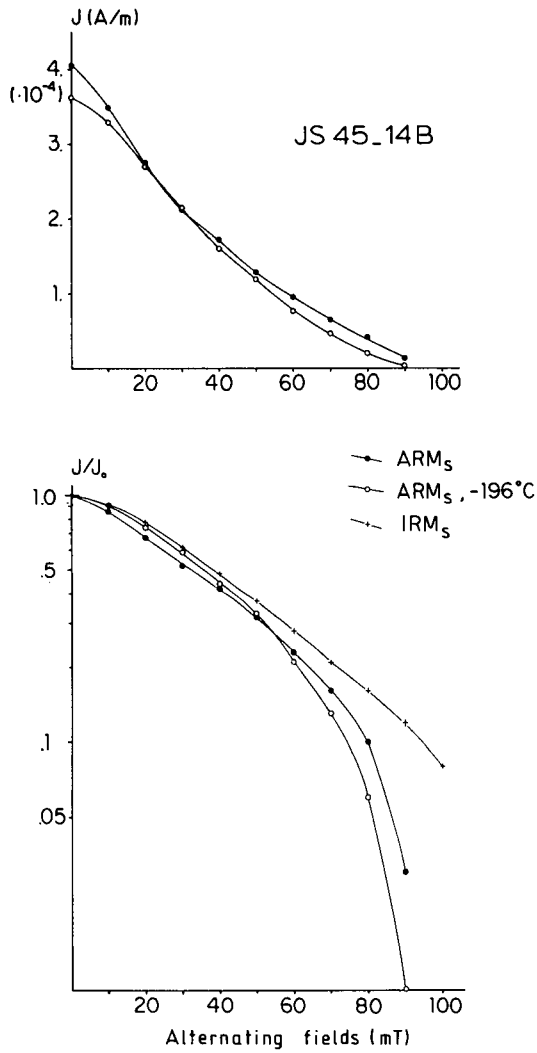


Figure 11. Alternating field demagnetization of ARM and IRM in a sample of blue limestone. The intensities of saturation ARM (ARMs) before and after cycling the sample to −196 °C are shown in the top plot. IRMs = 1.37 × 10^{-3} A/m.

It is evident from the AF demagnetization of NRM in this large number of blue limestone samples that the carriers of NRM in these samples have low to intermediate coercivity. IRM-acquisition experiments were also conducted with blue limestone samples, and highly uniform results, in which all samples reached saturation IRM in applied fields less than 0.3 T (Fig. 4b), confirm the low coercivity of magnetic carriers in these rocks. IRM-acquisition characteristics in samples from different sites (see Fig. 4) furthermore indicate similarity in magnetic carriers present in the blue limestone samples examined. Variation in saturation IRM intensity between samples therefore probably represents differing quantities of magnetic material present in different samples. Although blocking temperatures for the NRM in blue limestones could not be determined precisely from the results for the few samples treated using thermal methods, thermal demagnetization of saturation IRM was completed for many samples. These experiments show that IRM in all samples decayed uniformly with increasing temperature until all magnetization was removed by approximately 580 °C (Fig. 10), the Curie temperature of magnetite. Several magnetic properties thus indicate the presence of magnetite in the blue limestones, and it is concluded that it is the probable carrier of the observed NRM in these rocks.

Results from IRM-acquisition experiments and thermal demagnetization of IRMs that suggest no magnetic mineral phases other than magnetite in the blue limestones also indicate that standard methods may be used in experimental tests to investigate the domain state of the magnetite in these

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

samples (Merrill, 1970; Johnson and others, 1975). Blue limestone core samples thus were given an ARM, and its stability in progressive AF treatment and the intensity change produced by cycling samples to low temperature (−196 °C) were examined. Stepwise magnetization showed that ARMs was just achieved by 100 mT in most blue limestone samples studied. AF demagnetization also to 100 mT removed the ARMs given to samples (Fig. 11). Cycling of samples to low temperature had very little effect on either the intensity of the initial ARMs or its stability with respect to AF treatment (Fig. 11). This would indicate that relatively little multidomain magnetite is present in these samples. However, very similar AF stabilities for ARMs and IRMs in these samples (see Fig. 11) suggest that magnetite is probably not carried solely in single-domain magnetite grains. Pseudo-single-domain remanence carriers may be important in the Jura limestones, although this cannot be conclusively demonstrated from these simple tests.

IMPLICATIONS FOR DIAGENESIS

Geologic and paleomagnetic data for limestones from the Jura Mountain region suggest that the tan coloration of these limestones is a diagenetic product. In particular, paleomagnetic data demonstrate that goethite, which imparts this color, formed during late Tertiary or Quaternary times. A number of diagenetic modes of origin for this goethite are possible but field relations between tan and blue limestones in the Jura indicate that tan coloration results from diagenetic alteration of pyrite contained in the blue limestones. Pyrite occurs only in samples of blue limestone, but in tan limestones goethite pseudomorphs after pyrite occur. This supports an interpretation that the blue limestones served as a parent material in which pyrite was oxidized to goethite. The predominance of magnetite in the blue limestone and its subsidiary role to that of goethite in the tan limestones are consistent with this model. Conceivably, much of the original fine-grained magnetite in a parent material also would have been oxidized if conditions sufficient to oxidize pyrite were achieved. It is noted that the presence of water is required in such a model in order for oxidation of pyrite to take place and to allow for the transport of ferric iron which ultimately was reprecipitated as pervasive finely disseminated goethite in tan limestones. The timing of goethite formation suggests diagenesis coeval with uplift and subareal exposure of the Jurassic strata when meteoric water probably was available.

We suggest that blue-gray limestones in the Jura region are the best record of original lithologies that escaped diagenesis. The late Tertiary diagenetic process inferred for the tan limestones requires oxidation of iron-bearing minerals by fluids and therefore must be dependent upon relative permeabilities of the Jurassic strata. We suggest, therefore, that the preservation of blue limestones reflects low permeabilities in some strata. For example, limestones of the Terrain à Chailles are contained within relatively impermeable clay-rich strata that may have isolated them from oxidation by meteoric waters. The common occurrence of blue limestones in the Terrain à Chailles supports this interpretation.

In contrast to goethite in the tan limestones, the magnetite carrier in the blue limestones most likely was present at the time of deposition of the limestones, and therefore the characteristic magnetization in these rocks is of depositional or postdepositional or biogenic origin. Evidence for bioturbation of the limestones (massive textures, occasional preserved burrow structures), however, makes a depositional remanence less probable. A chemical origin for the characteristic magnetization, for example, at a time much later than that of deposition is considered to be highly unlikely because precipitation of authigenic magnetite in these rocks after burial is difficult to explain. Bioturbation of these sediments prior to lithification resulted, among other things, in preservation of very little organic matter, thus eliminating from these rocks an important source of reducing conditions that are necessary for the formation of magnetite. Fluids from an external source might have caused reducing conditions in these strata on a local scale, but due to the variable permeability apparent for some of these strata, reducing fluids could not have given rise to the remanence observed at sites widely distributed throughout the sampling region. A remagnetization of magnetite in the blue limestones by thermal processes is discounted because these rocks were never deeply buried and they were far removed from any possible thermal sources throughout their history. The almost exclusively normal polarity of characteristic remanence observed in the blue limestones from rock units ranging in age from lower Oxfordian through Kimmeridgian is consistent with the most recent paleomagnetic reversal time scale for that time span (Channell and others, 1982b), and this also supports an early acquisition for the characteristic magnetization in magnetite in these rocks. Likewise, reversed directions observed in younger Portlandian-age samples (site 7) are consistent with paleomagnetic reversals occurring in the latest Jurassic (Channell and others, 1982b; Ogg, 1981). On the other hand, we suggest that because normal and a few reversed directions from site 8 are in good agreement with characteristic directions from other sites, the reversed directions found here may represent a less well-documented reversal (event) in the Oxfordian.

In summary, we propose that NRM in magnetite was acquired early in the burial history of blue limestones. We do not suggest, however, that all magnetite contained in carbonate strata reflects a similar early origin. In other examples of this rock type, secondary magnetizations residing in magnetite are documented for which chemical or, less likely, thermal processes are responsible (Scotese and others, 1982). The presence of goethite, on the other hand, should signal possible exposure of its host rocks to oxidizing conditions, and perhaps to subareal exposure, subsequent to their formation. The magnetic oxide hematite may form in limestones under conditions similar to those inferred for the formation of goethite in the Jura region (Heller, 1978); however, paleomagnetic methods used to interpret the timing of hematite formation during early diagenesis in Italian Scaglia limestones (Channell and others, 1982a) indicate that alternate diagenetic processes can also be important in controlling the formation of hematite. At present, the geochemistry of Fe-minerals under near-surface conditions is not adequately known. Nonetheless, rock-magnetic and paleomagnetic methods such as those used in this study can provide important information on both the occurrence and time of formation of different Fe-oxides in carbonates. If the modes of origin and the stabilities of these and other Fe-minerals can be better understood, magnetic methods may be used in other investigations to infer geologic events or conditions that have affected carbonate rocks during their history.

A JURASSIC PALEOPOLE

The strata that have yielded site-mean poles for the blue limestones (Table 2) are of slightly different ages (early to late Oxfordian) (see Fig. 2). However, there is no significant trend in variation of position of site-mean poles in blue limestones with age. If, therefore, apparent polar wander for western Europe did occur during the time of deposition of these rocks, scatter in observed characteristic directions for the limestones prevents its

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

TABLE 5. A COMPARISON OF MID-UPPER JURASSIC PALEOPOLES FOR EUROPE, NORTH AMERICA, AND MOROCCO; TECTONIC ROTATIONS ARE MADE WITH EUROPE HELD FIXED

References	Rotation pole AFR/NAM	Rotation pole NAM/EUR	Rotation pole AFR/EUR†	Pole NAM[1] after rotation	Pole AFR[2] after rotation	Mean pole§
						K (N=4)
Pitman and Talwani, 1972	63.0N, −18.0E −68.0	88.4N, 27.4E 38.0	38.7N, −0.7E −37.2	68.0N, 179.6E	76.7N, 196.0E	74.7N, 161.5E 64.1
Le Pichon and others, 1977	67.0N, −13.7E −63.5	85.0N, 129.9E 26.7	50.6N, −2.1E −41.2	68.0N, 161.3E	74.0N, 194.6E	73.7N, 156.5E 73.5
Kristoffersen, 1978	63.0N, −18.0E −68.0*	71.2N, 148.4E 23.1	46.0N, −5.8E −54.1	66.4N, 144.6E	76.3N, 145.8E	72.6N, 140.5E 166.6

[1]Pole before rotation is mean [67.2N, 139.9E; A_{95} = 10.4] of four individual investigation poles from North America : 68.4N,145.6E (McCabe and others, 1982); 67.N,110.E (Steiner, 1978); 68.N,162.E and 61.N,142.E (both from Steiner and Helsley, 1975).
[2]Pole before rotation is 53.N,261.5E (Hailwood and Mitchell, 1971).
†Rotation poles AFR/EUR for three reconstructions were calculated from rotation poles AFR/NAM, NAM/EUR reported by the authors cited.
§Mean of European poles at 77.7N,148.4E (this study) and 69.N,128.E (Heller, 1977), and poles NAM and AFR after rotation; calculated for each of three tectonic models.
*Kristoffersen (1978) presented a rotation pole for NAM/EUR only; the rotation pole AFR/NAM used here is from Pitman and Talwani (1972).

resolution. Similarly, no significant difference between pole positions for sites in folded beds and those for sites in flat-lying and very nearly flat-lying beds is seen. Relative rotations between sites thus are not detected. The scatter in characteristic directions is most likely due to nondipole variations in the Late Jurassic paleomagnetic field, or it might have been introduced as small measurement error. If the variation in site-mean directions is predominantly a result of (time-varying) nondipole components in the paleomagnetic field, the directions for the different-age, blue-limestone strata should be scattered (in a Fisher distribution; Fisher, 1953) about the ancient dipole-field direction.

A small undetected systematic error in laboratory measurement of samples would have a negligible effect on a mean direction calculated for such a large number of samples. The mean direction or mean pole for blue limestone samples (Table 3) thus is considered to be representative of an average Late Jurassic (Oxfordian) paleomagnetic-dipole field with respect to western Europe.

In similar limestones of Late Jurassic age from southern Germany, Heller (1978) obtained a pole position at 69N128E, which lies some 10° from the pole found in this paper at 78N148E (Table 5). Characteristic directions in magnetite are also present in the rocks studied by Heller, and they gave a positive fold test indicating that remanence was acquired prior to Miocene faulting. This is important because the directions found by Heller are close to present-day field directions. The statistically significant fold test for characteristic magnetizations in the Jura limestones confirms a high-latitude position for the Upper Jurassic European pole.

The mean pole for the Jura limestones also can be compared with the few paleopole data available for Middle to Upper Jurassic rocks for other continents. Such a comparison is made in Table 5 and Figure 12 between a mean Upper Jurassic pole from North America (see Table 5), a single Upper Jurassic pole for Africa (Morocco) (Hailwood and Mitchell, 1971), and both the Jura limestone pole and the pole determined by Heller. In this comparison, the poles from Europe are held fixed and the poles from North America and Morocco are rotated according to published tectonic rotation poles that have been used to reconstruct the configuration of the Atlantic-bordering continents during this time. The rotation parameters for the motion between North America and Africa were derived by fitting sea-floor magnetic anomalies of Middle to Late Jurassic age (Quiet Zone boundaries) from either side of the mid-Atlantic ridge (Pitman and Talwani, 1972) and from a combination of magnetic anomaly data and consideration of structural features preserved along the margins of the Atlantic that are thought to have been developed during continental breakup (Le Pichon and others, 1977). The rotation parameters for motion between North America and Europe are those for the fit by Bullard

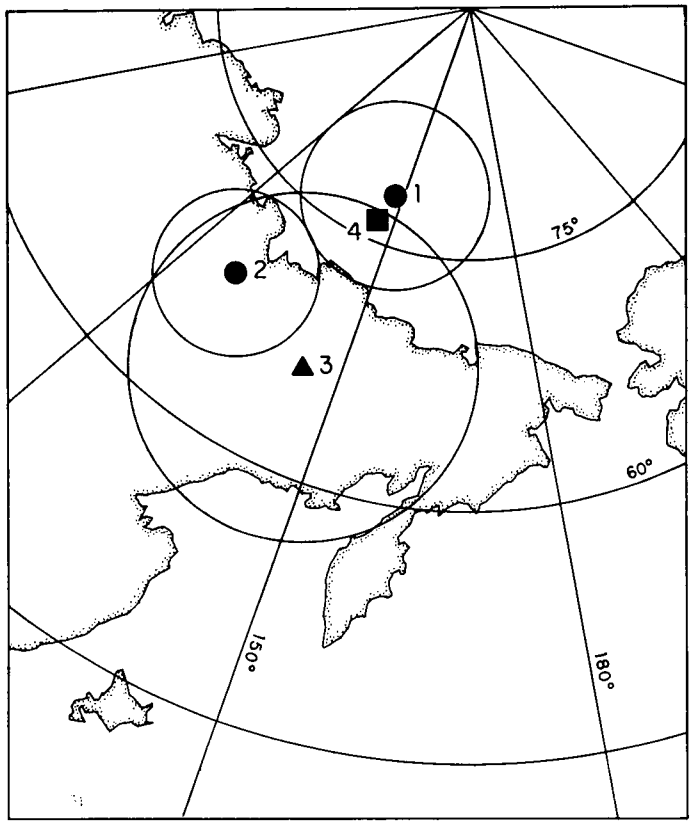


Figure 12. Mid-Upper Jurassic paleopoles for western Europe, from the Jura limestones (1) and from Heller (1977) (2), compared with a mean Upper Jurassic pole for North America (3) (see Table 5) and a single Upper Jurassic paleopole from Africa (4). The North American and African poles have been rotated using the reconstruction of Kristoffersen (1978), while holding Europe fixed. Circles for European and North American poles are circles of 95% confidence (A_{95}; Fisher, 1953); the much larger circle about the Moroccan pole (A_{95} = 24) has not been drawn.

and others (1965) as used by Pitman and Talwani (1972), those determined from a reinterpretation of older magnetic anomalies in the North Atlantic (Kristoffersen, 1978), and rotation parameters determined from consideration of the spreading histories of oceanic regions between Greenland, North America, and Europe (Le Pichon and others, 1977). Al-

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

though there are uncertainties in the paleomagnetic pole positions (A_{95} = 24 for the Moroccan pole is notably very large), and in the rotation parameters used for reconstruction, it can be seen from Table 5 that the paleomagnetic poles used are in reasonably good agreement after the rotations are performed. The few paleomagnetic data used, plus the uncertainty in the rotations applied to these poles, do not allow rigorous determination of a preferred reconstruction; however, we note that the relative positions for North America and Europe derived by Kristoffersen (1978) in conjunction with a position for Africa determined by either Pitman and Talwani (1972) or Le Pichon and others (1977) yield a mean Upper Jurassic pole position with high values for k (167 and 116, respectively).

CONCLUSIONS

From application of a fold test for the age of acquisition of observed magnetizations in fine-grained limestones from the Jura Mountain region, it has been learned that goethite, which gives a tan color to a large proportion of these strata, had a diagenetic origin sometime after the Miocene (but probably pre-Brunhes). Blue-gray limestones found very infrequently in the Jura region, on the other hand, contain very little goethite and apparently have escaped the extensive diagenetic modifications prevalent in other rocks of the same age. Magnetite in blue limestones has been present since before deformation of these strata and it is probably primary.

This study of limestones from the Jura Mountain region demonstrates the usefulness of paleomagnetic methods for dating diagenetic events in limestones and other carbonate rocks. In some examples, paleomagnetic methods may reveal information about the geologic processes that operated to produce secondary magnetizations in such rocks. In the Jura region, for instance, goethite in limestones appears to reflect the introduction of meteoric water into contact with the host rocks. It thus may be possible to use paleomagnetic methods to investigate similar significant events elsewhere.

A mean Upper Jurassic paleopole for western Europe is well determined from characteristic directions carried in magnetite in blue limestones from the Jura region. The Jura pole and one other pole for Jurassic rocks from western Europe, determined by Heller (1978), are in good agreement. A comparison of these poles with the few paleopoles of approximately the same age available from other continents has shown the potential merit of recent interpretations of the reconstruction of Atlantic-bordering continents during this time. Polarities observed for characteristic directions in the Jurassic limestones corroborate a proposed long period of uniform polarity during most of the Oxfordian and Kimmeridgian, but evidence also exists for a reversal event during the Oxfordian.

ACKNOWLEDGMENTS

We thank Dr. R. Gygi for valuable advice on the occurrence of suitable outcrops, and Alfred Bürgin and Chris Lusser of the University of Basel for their extensive and invaluable help in the field. K. C. Lohmann provided many helpful comments on the manuscript. Research was supported by the Institut für Geophysik of the ETH-Zurich and by the National Science Foundation, Division of Earth Sciences, grant EAR 79-02690.

REFERENCES CITED

Bolliger, W., and Burri, P., 1970, Sedimentologie von shelf-carbonaten und Beckenablagerungen in Oxfordien des centralen Schweizer Jura: Beitrage zur Geologie Karte Schweiz, Ser. 140.

Bullard, E., Everett, J. E., and Smith, A. G., 1965, The fit of the continents around the Atlantic: Royal Society of London Philosophical Transactions, ser. A, v. 258, p. 41–51.

Channell, J.E.T., Freeman, R., Heller, F., and Lowrie, W., 1982a, Timing of diagenetic hematite growth in red pelagic limestones from Gubbio (Italy): Earth and Planetary Science Letters, v. 58, p. 189–201.

Channell, J.E.T., Ogg, J. G., and Lowrie, W., 1982b, Geomagnetic polarity in the Early Cretaceous and Jurassic: Royal Society of London Philosophical Transactions, ser. A, v. 306, n. 1492, p. 137–146.

Dunlop, D. J., 1972, Magnetic mineralogy of unheated and heated red sediments by coercivity spectrum analysis: Royal Astronomical Society Geophysical Journal, v. 27, p. 37–55.

Fisher, R. A., 1953, Dispersion on a sphere: Royal Society of London Proceedings, ser. A, v. 217, p. 295–305.

French, R. B., 1976, Lower Paleozoic paleomagnetism of the North American Craton [Ph.D. thesis]: Ann Arbor, Michigan, University of Michigan, 159 p.

Gignoux, M., 1955, Stratigraphic geology: San Francisco, W. H. Freeman and Co., Inc., 682 p.

Goree, W. S., and Fuller, M. D., 1976, Magnetometers using RF-driven squids and their applications in rock magnetism and paleomagnetism: Reviews of Geophysics and Space Physics, v. 14, p. 591–608.

Graham, J. W., 1949, The stability and significance of magnetism in sedimentary rocks: Journal of Geophysical Research, v. 54, p. 131–167.

Gygi, R., 1969, Zur stratigraphie der Oxford-Stufe (oberes Jura-System) der Nordschweiz und des suddeutschen Grenzgebietes: Beitrage zur Geologie Karte Schweiz, New Series 136, No. 11.

Hailwood, E. A., and Mitchell, J. G., 1971, Paleomagnetic and radiometric dating results from Jurassic intrusions in south Morocco: Royal Astronomical Society Geophysical Journal, v. 24, p. 351–364.

Heller, F.A.R., 1977, Paleomagnetism of Upper Jurassic limestones from southern Germany: Journal of Geophysics, v. 44, p. 525–543.

——1978, Die magnetischen Eigenschaften der Oberjurakalke Suddeutschlands [Habilitationsschrift]: Zurich, ETH, 75 p.

Johnson, H. P., Lowrie, W., and Kent, D. V., 1975, Stability of anhysteretic remanent magnetization in fine and coarse magnetite and maghemite particles: Royal Astronomical Society, Geophysical Journal, v. 41, p. 1–10.

Kobayashi, K., and Fuller, M. D., 1968, Stable remanence and memory of multidomain materials with special reference to magnetite: Philosophical Magazine, v. 18, p. 601–624.

Kristoffersen, Y., 1978, Sea-floor spreading and the early opening of the North Atlantic: Earth and Planetary Science Letters, v. 38, p. 273–290.

Laubscher, H. P., 1961, Die Fernschubhypothese der Jurafaltung: Eclogae Geologicae Helvetiae, v. 54, no. 1, p. 221–282. (with English abstract).

——1982, The 3D propagation of decollement in the Jura, *in* McClay, K. R., and Price, N. J., eds., Thrust and nappe tectonics: Geological Society of London Special Publication No. 9, p. 311–318.

Le Pichon, X., Sibuet, J-C., and Francheteau, J., 1977, The fit of the continents around the Atlantic Ocean: Tectonophysics, v. 38, p. 169–209.

Levi, S., and Merrill, R. T., 1976, A comparison of ARM and TRM in magnetite: Earth and Planetary Science Letters, v. 32, p. 171–184.

Lowrie, W., and Heller, F.A.R., 1982, Magnetic properties of marine limestones: Reviews of Geophysics and Space Physics, v. 20 (2), p. 171–192.

Lowrie, W., Channell, J.E.T., and Alvarez, W., 1980, A review of magnetic stratigraphy investigations in Cretaceous pelagic carbonate rocks: Journal of Geophysical Research, v. 85, no. B7, p. 3597–3605.

McCabe, C., Van der Voo, R., and Wilkinson, B. H., 1982, Paleomagnetic and rock magnetic results from the Twin Creek Formation (Middle Jurassic), Wyoming: Earth and Planetary Science Letters, v. 60, p. 140–146

McElhinny, M. W., 1964, Statistical significance of the fold test in paleomagnetism: Royal Astronomical Society Geophysical Journal, v. 8, p. 338–340.

Merrill, R. T., 1970, Low-temperature treatments of magnetite and magnetite bearing rocks: Journal of Geophysical Research, v. 75, p. 3343–3349.

Ogg, J. G., 1981, Sedimentology and paleomagnetism of Jurassic pelagic limestones ("ammonitico rosso facies") [Ph.D. thesis]: San Diego, California, University of California, 203 p.

Ozima, M., Ozima, M., and Akimoto, 1964, Low temperature characteristics of remanent magnetization of magnetite: Journal of Geomagnetism and Geoelectricity, v. 16, p. 165–177.

Pierce, W. G., 1966, Jura tectonics as a decollement: Geological Society of America Bulletin, v. 77, p. 1265–1276.

Pitman, W. C., and Talwani, M., 1972, Sea-floor spreading in the North Atlantic: Geological Society of America Bulletin, v. 83, p. 617–649.

Rutten, M. G., 1969, The geology of western Europe: Amsterdam, Elsevier, 520 p.

Schott, J-J., Montigny, R., and Thuizat, R., 1981, Paleomagnetism and potassium-argon age of the Messejana Dike (Portugal and Spain): Angular limitation to the rotation of the Iberian Peninsula since the Middle Jurassic: Earth and Planetary Science Letters, v. 53, p. 457–470.

Scotese, C., Van der Voo, R., and McCabe, C., 1982, Paleomagnetism of the Upper Silurian and Lower Devonian carbonates of New York State: Evidence of secondary magnetizations residing in magnetite: Physics of the Earth and Planetary Interiors, v. 30, no. 4, p. 385–395.

Steiner, M. B., 1978, Magnetic polarity during the Middle Jurassic as recorded in the Summerville and Curtis Formations: Earth and Planetary Science Letters, v. 38, p. 331–345.

Steiner, M. B., and Helsley, C. E., 1975, Reversal pattern and apparent polar wander for the Late Jurassic: Geological Society of America Bulletin, v. 86, p. 1537–1543.

Van Eysinga, F.W.B., 1975, Geological time table: Amsterdam, Elsevier.

Wilson, J. L., 1975, Carbonate facies in geologic history: New York, Springer-Verlag, 471 p.

Zijderveld, J.D.A., 1967, AC demagnetization of rocks: Analysis of results, *in* Collinson, D. W., Creer, K. M., and Runcorn, S. K., eds., Methods in paleomagnetism: Amsterdam, Elsevier, p. 254–286.

MANUSCRIPT RECEIVED BY THE SOCIETY APRIL 9, 1982
REVISED MANUSCRIPT RECEIVED APRIL 16, 1983
MANUSCRIPT ACCEPTED MAY 2, 1983
CONTRIBUTION NO. 365 OF THE INSTITUT FÜR GEOPHYSIK OF THE ETH